When Machine Learning Beats Persistence: A Leakage-Free Streamflow and Drought Benchmark for the Tropical Andes

Richar Andre Vilca-Solorzano, Renato Quispe-Vargas and Dina Maribel Yana-Yucra

Abstract Machine-learning studies of Andean streamflow rarely compare their models with persistence, and their validation designs can leak information from the test period. We built an open benchmark of 19 long-record gauges from the Andean Glacierized Catchment dataset in three regions: the Altiplano, including Puente Isla Cabanillas in the Lake Titicaca basin, the Cordillera Blanca, and the western and southern Andes. Daily discharge was combined with ERA5 and MERRA-2 forcing and GloFAS v4 simulated discharge. Six learners were compared with persistence and climatology at 1-, 3- and 7-day leads under a calendar-continuous, embargoed protocol tuned on validation data only. Uncertainty was estimated with block-bootstrap intervals and Diebold–Mariano tests. The learners beat persistence at 15 of 19 gauges at one day and at 18 of 19 at seven days (median skill 0.101). Lower error did not improve drought detection: the median critical success index of drought days was 0.43 for persistence and 0.30 for the learners. Drought synchrony decayed over about 582 km, and skill depended on hydroclimatic regime rather than location.

Key words: Tropical Andes · Altiplano · Streamflow forecasting · Hydrological drought · Machine learning · Data leakage

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1 Introduction

Rivers of the tropical Andes supply irrigation, hydropower and drinking water, but their flow follows a strong seasonal cycle and is exposed to glacier retreat and recurrent drought [34]. On the Altiplano, rivers such as the Coata–Cabanillas feed Lake Titicaca and depend on a short austral-summer wet season [11]. Hydrological droughts in the Titicaca system are projected to become longer and more frequent [36]. Short-range discharge forecasts and early drought signals are therefore useful for water management. Gauges, however, are sparse and their records contain long gaps.

Machine learning (ML) is now widely used to forecast streamflow [15,24]. At daily leads, discharge is so autocorrelated that the persistence forecast $\hat{Q}_{t+h} = Q_t$ is hard to beat [26]. Reported gains can be inflated in three ways: by comparing models with weak references, by shuffled cross-validation that leaks temporal information [2, 33], and by giving the baseline and the learners different information at the time of issue. An earlier version of our own pipeline made the third error. The learners saw discharge only up to $t-1$, persistence used Q_t, and calendar gaps were treated as consecutive days. Correcting both errors reversed the one-day conclusion.

The aim of this study was to measure, under a protocol without leakage, how much ML adds to persistence for 1- to 7-day streamflow and drought forecasts in the Andes. The paper makes four contributions. First, it assembles an open benchmark of 19 gauges in three hydroclimatic regions. Second, it defines a leakage-free protocol that is checked by automated tests. Third, it reports skill against persistence with confidence intervals at three leads and compares it with earlier Andean studies. Fourth, it scores drought-day detection directly and describes the spatial coherence of drought and skill. Section 2 reviews related work, Sect. 3 describes the data and methods, Sect. 4 presents the results, Sect. 5 discusses them together with the limitations, and Sect. 6 concludes.

2 Related Work

Data-driven streamflow forecasting in the Andes. Random forests and LSTM networks have been used for hourly runoff forecasting in Ecuadorian Andean catchments [5, 20, 22]. In Chile, LSTM networks forced by weather forecasts supported flood warning in the Maipo basin [10], and random forests predicted monthly snowmelt-driven flows in the Elqui basin [25]. Support-vector regression was tested for monthly runoff in the Central Andes of Argentina [30]. In Peru, artificial neural networks forecast monthly flows in the Crisnejas basin [32], k-nearest neighbours forecast daily flows of the Ramis River on the Altiplano [19], and ensemble data assimilation improved hourly forecasts in the Vilcanota basin [17]. On the Altiplano, gridded precipitation agrees well with gauges but can still degrade simulated streamflow [18]. Large-sample studies with long short-term memory (LSTM) networks set the reference for data-driven hydrology [15, 24]. Table 1 summarizes these Andean

Table 1 Earlier data-driven streamflow studies in the Andes compared with this study. Pers.: persistence reference; CI: uncertainty of skill reported. Scores as reported by the authors (validation or test).

Study	Region	Gauges	Step, lead	Method	Pers.	CI	Reported score
Muñoz et al. 2018 [22]	Ecuador	1	4–24 h	RF	no	no	NSE 0.76–0.38
Contreras et al. 2021 [5]	Ecuador	1	4–24 h	RF	no	no	R^2 0.69–0.29
Merizalde et al. 2023 [20]	Ecuador	1	1–11 h	LSTM	no	no	NSE 0.93–0.67
de la Fuente et al. 2019 [10]	Chile	9	hourly, 3 d	LSTM	no	no	NSE ≈0.93–0.99
Núñez et al. 2023 [25]	Chile	1	1–4 months	RF	no	no	R^2 0.85–0.73
Teverovsky et al. 2023 [30]	Argentina	1	1 month	SVR	no[a]	no	NSE up to 0.90
Vásquez-R. et al. 2023 [32]	Peru	1	monthly	ANN	no	no	r^2 0.81
Lujano et al. 2023 [19]	Peru (Altiplano)	1	1 day	k-NN	no	no	NSE 0.979
This study	Peru, Chile	19	1–7 days	6 learners	yes	yes	skill 0.101 at 7 days

[a]Compared with seasonal ARIMA.

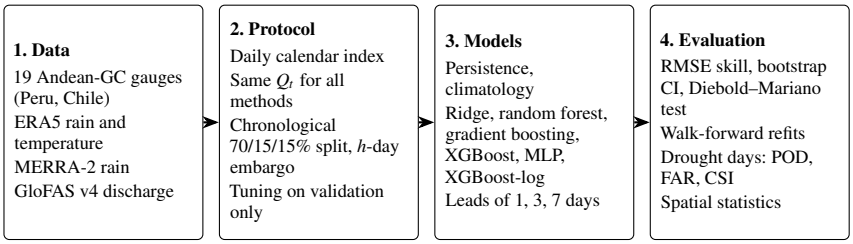


Fig. 1 Workflow of the benchmark. Code, tests and result tables are in the public repository.

studies. Seven of the eight used a single target gauge, none compared the models with persistence, and none reported an uncertainty estimate for the skill.

Validation of dependent data. Random k-fold cross-validation is optimistic for autocorrelated or spatially structured data [2]. For Peruvian agricultural survey data, blocked cross-validation showed that random folds overstated accuracy [33]. We apply the same principle in time, with chronological blocks, an embargo and walk-forward refits.

Hydrological drought. Hydrological drought is a below-normal availability of streamflow or storage [31]. It is identified with threshold-level methods [8] or with standardized streamflow indices (SSI), whose nonparametric versions avoid fitting a distribution [7]. The drought definition also changes forecast verification [28, 29], so we score drought days explicitly rather than infer them from the error of the discharge forecast.

3 Data and Methods

Figure 1 summarizes the workflow. **Discharge.** We used quality-controlled (QC) daily discharge from the Andean-GC dataset [1] (CC BY 4.0). We kept every Peruvian gauge, operated by the National Meteorological and Hydrological Service

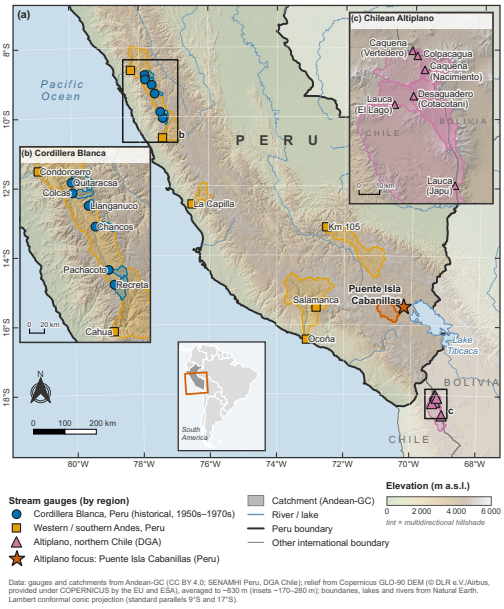


Fig. 2 The 19 benchmark gauges and their catchments, by region. The Altiplano focus gauge, Puente Isla Cabanillas, is highlighted. Insets show the Cordillera Blanca and the Chilean Altiplano.

(SENAMHI), and every Altiplano gauge of the Lauca and Caquena headwaters in Chile, operated by the General Directorate of Water (DGA), with at least 5000 QC days. This gave 19 gauges: 7 on the Altiplano, 6 glacier-fed gauges in the Cordillera Blanca, and 6 in the western and southern Andes (Fig. 2). The focus gauge was Puente Isla Cabanillas (2843 km^2, mean elevation 4471 m), which drains the Coata–Cabanillas system into Lake Titicaca. The Cordillera Blanca records cover 1952–1974; the other records span 1950–2024. Multi-year gaps were kept as missing values on a daily calendar.

Forcing and simulated discharge. Daily precipitation and 2-m temperature were taken from ERA5 [13] at each catchment centroid through the Open-Meteo API [35]. A second precipitation estimate was NASA POWER `PRECTOTCORR`, based on MERRA-2 and available from 1981. GloFAS v4 reanalysis discharge (from 1984) was read on a 3×3 window of 0.05° cells around each gauge. The cell with the highest Kling–Gupta efficiency (KGE) over the fitting period was used, so the test period played no part in this choice.

Forecast set-up. A forecast issued on day t predicts Q_{t+h}, $h \in \{1,3,7\}$ days, from 27 predictors available at t. These were discharge lags Q_{t-k} ($k = 0,1,2,3,7,14,30$), the one-day change, 7-, 14- and 30-day rolling means and standard deviations, precipitation lags (0–7 days) and 7-, 30- and 90-day totals, temperature means, and the day of year as sine and cosine. All operations used a daily calendar index, so a

lag of k days is exactly k days, and samples with a predictor or target in a gap were dropped. Persistence used the same Q_t that the learners saw.

Methods compared. The two references were persistence and a day-of-year climatology. The six learners were Ridge regression [14], random forest [3], histogram gradient boosting [9], XGBoost [4], a two-layer multilayer perceptron (MLP), and an XGBoost trained on the change in $\log(1+Q)$ to favour low flows (XGBoost-log). For each learner, a small grid of hyper-parameters was searched together with the target form, either the level Q_{t+h} or the change $Q_{t+h}-Q_t$.

Validation protocol. Each record was split chronologically into 70% training, 15% validation and 15% test. The last h samples of the training and validation blocks were removed (embargo), so no target overlapped the next block. The configuration with the lowest validation root-mean-square error (RMSE) was refit on training plus validation and scored once on the test block. "The learner" of a gauge denotes this validation-selected model, so the test set was never used to choose models. Ridge was also run in an expanding-window walk-forward with three folds at every gauge, with hyper-parameters re-selected in each fold.

Scores. We report RMSE, NSE [23], KGE [12] and the skill against persistence, $S = 1 - \mathrm{RMSE}_{\mathrm{model}}/\mathrm{RMSE}_{\mathrm{pers}}$. The 95% confidence intervals (CIs) of S come from a paired moving-block bootstrap (1000 replicates, 30-day blocks) [16]. Equal accuracy with persistence was tested with the Diebold–Mariano test with small-sample correction [6].

Drought detection and spatial statistics. A day was in hydrological drought when Q fell below the monthly 20th percentile of the fitting period [8, 31]. Forecast and observed drought states at $t+h$ were compared through the probability of detection (POD), false-alarm ratio (FAR) and critical success index (CSI) [27]. A nonparametric SSI-30 was computed from 30-day mean discharge ranked within each calendar month [7]. Drought synchrony was measured as the Spearman correlation of monthly SSI-30 between gauge pairs with at least 60 common months, and fitted with $\rho(d) = c\,e^{-d/d_0}$ against distance d; the CI of d_0 came from a gauge-level bootstrap. Spatial clustering of the 7-day skill was tested with Moran's I [21], and its relation to catchment attributes with Spearman correlations and a Holm correction. In a data-fusion test, Ridge and XGBoost were refit at the gauges covered by MERRA-2 and GloFAS with and without these two sources. The pipeline runs on a laptop in about 4 min, and all numbers in the text are generated from the result tables.

4 Results

Focus gauge. At Cabanillas (Table 2; test period 2021-09-22 to 2024-12-24), the validation-selected random forest reached NSE = 0.922 at one day, against 0.902 for persistence. This is a skill of 0.110 (95% CI 0.064 to 0.144; Diebold–Mariano p <0.001), the reverse of the result of our earlier, leaky pipeline. At seven days, persistence fell to NSE = 0.461. Ridge on the change target gained 0.115 with a CI above zero, and its walk-forward folds gave skills of 0.166, 0.114, 0.114. The learner

Table 2 Test scores at Puente Isla Cabanillas. Skill is relative to persistence, with 95% CI. †Learner selected on validation. CSI: drought-day detection at seven days.

	1-day lead		7-day lead		
Method	NSE	Skill [95% CI]	NSE	Skill [95% CI]	CSI
Persistence	0.902	–	0.461	–	0.61
Ridge	0.917	0.081 [0.03, 0.12]	0.578	0.115 [0.03, 0.21]	0.47
Random forest	0.922	0.110† [0.06, 0.14]	0.528	0.064 [−0.02, 0.14]	0.59
Hist. gradient boosting	0.916	0.074 [0.01, 0.14]	0.437	−0.022 [−0.16, 0.12]	0.18
XGBoost	0.915	0.071 [−0.01, 0.13]	0.407	−0.049 [−0.19, 0.09]	0.13
MLP	0.927	0.141 [0.09, 0.18]	0.203	−0.216 [−0.48, −0.01]	0.36
XGBoost-log	0.916	0.077 [0.00, 0.13]	0.502	0.038† [−0.08, 0.18]	0.56

selected on validation at this lead (XGBoost-log) had a smaller skill of 0.038, not significant.

All gauges. The learner beat persistence at 15/19 gauges at one day, 15/19 at three days and 18/19 at seven days, and its CI excluded zero at 9, 10 and 12 gauges (Fig. 3). The median skill grew with lead time, from 0.046 to 0.080 and 0.101, because persistence degraded faster than the learners. Between one and seven days, the median skill rose from 0.032 to 0.118 in the Cordillera Blanca and from 0.047 to 0.110 in the western and southern Andes, but stayed flat on the Altiplano (0.081 to 0.057). Ridge alone was skilful at 18/19 gauges at seven days. In the walk-forward check, 53 of 57 gauge–fold combinations had positive seven-day Ridge skill (median 0.103). Adding MERRA-2 and GloFAS to ERA5 changed the seven-day skill by a median of -0.004 for Ridge at the 7 gauges where they overlap the record, and the raw GloFAS simulation had a median NSE of -0.10.

Drought detection. Lower RMSE did not give better detection of drought days (Fig. 4). At seven days, the median CSI was 0.43 for persistence and 0.30 for the learner, which beat persistence at only 2/19 gauges. XGBoost-log did not close the gap (median CSI 0.33). The learners raised fewer false alarms than persistence (median FAR 0.35 vs. 0.39) but missed more drought days (POD 0.42 vs. 0.60).

Spatial statistics. The monthly SSI-30 correlation of 104 gauge pairs decreased with distance (Spearman -0.56, $p < 0.001$), with a decorrelation distance $d_0 = 582$ km (95% CI 330–1069 km; Fig. 5b). Pairs within a region correlated more strongly than pairs across regions (median ρ 0.28 vs. 0.09). Skill showed no spatial clustering (Moran's $I = 0.018$, $p = 0.374$; Fig. 5a). It decreased with snow fraction (ρ =-0.57, permutation $p = 0.010$, Holm-adjusted $p = 0.071$), precipitation concentration and aridity, the conditions of the Altiplano headwaters.

5 Discussion

The largest change in our conclusions came from the protocol, not from the choice of model. When persistence and the learners received the same information and lags followed the calendar, a one-day loss at Cabanillas became a significant gain. The

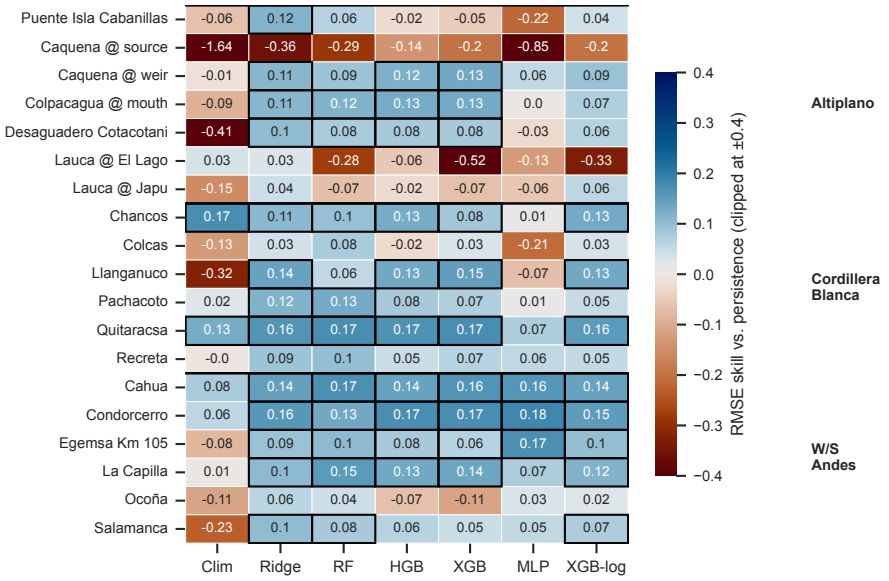


Fig. 3 Seven-day RMSE skill against persistence for every gauge and method. Boxed cells: 95% block-bootstrap interval above zero. Gauges are grouped by region (top to bottom: Altiplano, Cordillera Blanca, western and southern Andes).

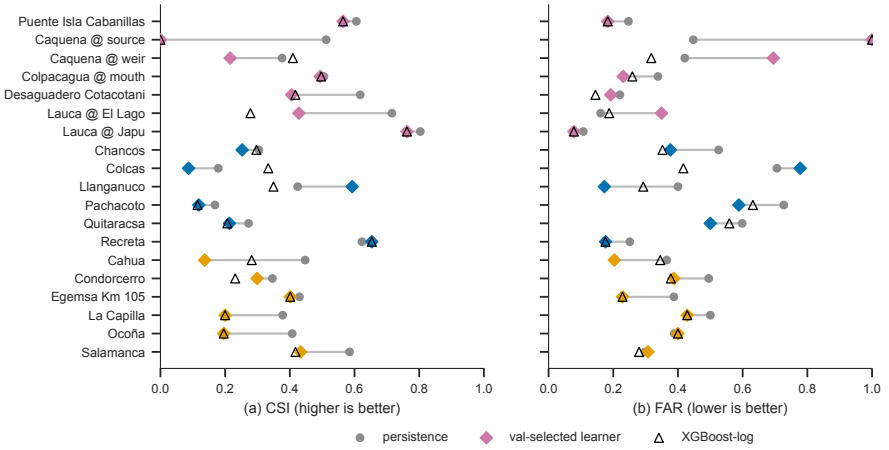


Fig. 4 Seven-day detection of drought days (threshold: monthly 20th percentile of the fitting period). Left: CSI; right: false-alarm ratio.

gains are modest (median RMSE reduction of about 5–10%) but consistent across gauges and walk-forward folds. Linear models on the change target were as good as tree ensembles, in line with large-sample evidence that careful baselines are hard to beat at short leads [24, 26].

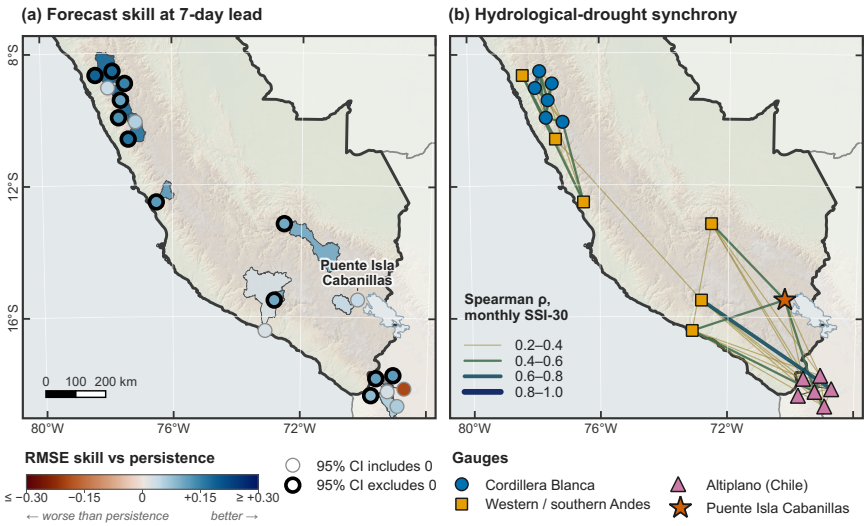


Fig. 5 (a) Seven-day skill of the validation-selected learner by catchment (bold outline: 95% CI above zero). (b) Drought-synchrony network: monthly SSI-30 Spearman correlation between gauge pairs with at least 60 common months.

These results help to read the earlier Andean studies (Table 1). Their scores are high, often NSE above 0.9 at short leads, but none was compared with persistence. At Cabanillas, persistence alone reached NSE = 0.902 at one day, and the best learner 0.922. For the Ramis River, also on the Altiplano, a k-NN model with only the previous day's discharge reached NSE = 0.965, against 0.979 for the best model [19]. In both rivers most of the NSE comes from persistence. A high NSE is therefore not evidence of forecast skill at daily leads, and we suggest that Andean studies report skill against persistence with a confidence interval.

The Altiplano was the hardest region. Its headwaters combine short convective wet-season pulses with long baseflow recessions [11], and a 0.25° reanalysis represents this rainfall only in part [18]. Recent observed discharge carried far more short-range information than MERRA-2 or GloFAS. Drought detection behaved differently from RMSE because models trained on squared error regress towards the conditional mean and smooth the recession limb, close to the drought threshold. Drought warnings should therefore be trained and verified on the drought decision itself [29]. Because drought decorrelated over a few hundred kilometres, one gauge cannot represent a whole region.

The study has several limitations. ERA5 is coarse for steep catchments and has a wet bias over the Andes. The Cordillera Blanca records predate 1975, so the fusion test covered only 7 gauges. A single validation block can be unstable under non-stationarity such as glacier retreat, as the seven-day selection at Cabanillas showed. Predictors used observed forcing up to t, not operational weather forecasts, so the

results are an upper bound for skill driven by forcing. Finally, the benchmark has 7 Altiplano gauges, and Bolivian records were not available.

6 Conclusion

We released an open, leakage-free benchmark for 1- to 7-day streamflow and drought forecasting at 19 Andean gauges. Under this protocol, the learners beat persistence at most gauges, at 18/19 gauges at seven days (median skill 0.101) and also at one day at Cabanillas. The gains in RMSE did not carry over to drought-day detection, where persistence remained a strong reference. Drought synchrony decayed over about 582 km, and skill depended on hydroclimatic regime rather than location. Future work will add drought-aware losses, operational weather forecasts and more Altiplano gauges.

Data and code availability

Discharge data: Andean-GC, doi:10.5281/zenodo.18035801 (CC BY 4.0). ERA5 and GloFAS were retrieved through Open-Meteo [35] and MERRA-2 precipitation through NASA POWER. Code, leakage tests, the gauge list, result tables and supplementary figures (hydrographs, lead-time skill, fusion test, walk-forward and correlogram) are available at `https://github.com/Andre031222/streamflow-drought-altiplano`.

Declarations

Author contributions. R. A. Vilca-Solorzano conceived the study, wrote the software, performed the analysis and wrote the manuscript. R. Quispe-Vargas validated the results and revised the manuscript. D. M. Yana-Yucra supervised the study, contributed to the methods and revised the manuscript. All authors approved the final version. **Competing interests.** None. **Use of AI.** Generative AI was used to improve the language; the authors reviewed the text and take full responsibility for it. We thank the providers of the open data listed above.

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